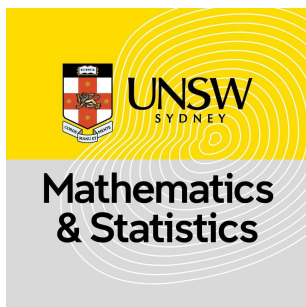




MATH3871/MATH5960 Bayesian Inference and Computation

Term 3, 2026

Tutorial Problems: Week 4

1. For the conjugate normal model, if $x_1, \dots, x_n \sim N(\theta, \tau^{-1})$ for known τ , then taking a $N(b, c^{-1})$ prior for θ gives the posterior

$$N\left(\frac{cb + n\tau\bar{x}}{c + n\tau}, \frac{1}{c + n\tau}\right).$$

Find the point estimate of θ that minimises each of quadratic, 0-1, absolute error, and linear loss.

2. A sample of 6 crop heights are recorded as

5.3, 5.6, 5.9, 6.1, 6.2, 6.5,

with $\bar{x} = 5.93$, and $s \approx 0.43$. Assuming a normal model, $x \sim N(\theta, \sigma^2)$, with a conjugate prior $\theta \sim N(b, d^2)$, the posterior is given by

$$\theta|x \sim N\left(\frac{\frac{b}{d^2} + \frac{n\bar{x}}{\sigma^2}}{\frac{1}{d^2} + \frac{n}{\sigma^2}}, \frac{1}{\frac{1}{d^2} + \frac{n}{\sigma^2}}\right).$$

Suppose that we wish to use a diffuse prior, with $d^2 \rightarrow \infty$. Calculate 90% HDR (high density) regions for the posterior population mean, θ , assuming that

- (a) the population variance is $\sigma^2 = 1$.
 - (b) the population variance is unknown (use Jeffrey's prior $\pi(\sigma^2) \propto 1/\sigma^2$). *Hint: you may find the worked example on Lectures Week 3, slides 17–22 useful.*
3. The weights of items from a particular production process are independently and identically distributed, each with a $N(\theta, 4)$ distribution. The production manager believes that θ varies from batch to batch according to a $N(110, 0.4)$ distribution. Items produced in the same batch will have the same value of θ . A sample batch of 5 items is randomly selected, yielding the measurements

108.0, 109.0, 107.4, 109.6, 112.0,

for which $\bar{x} = 109.2$.

- (a) Derive the posterior distribution for θ .
- (b) Find the predictive distribution for the weight of one further item from the batch.
- (c) Find the predictive distribution of the sample mean of the weights of a batch of m further items. *Hint: First write down the distribution of $\bar{y}_m = \sum_{i=1}^m y_i$, and then take the expectation of this with respect to the posterior.*

(d) What happens in (c) as $m \rightarrow \infty$?

4. The distribution of flaws along the length of an artificial fibre follows a Poisson process, so that the number of flaws in a length ℓ of the fibre is $\text{Poisson}(\ell\theta)$. Suppose we adopt a conjugate $\text{Gamma}(p, q)$ prior for θ . The number of flaws obtained in 5 fibres of lengths $\ell = (10, 15, 25, 30, 40)$ meters respectively were $x = (3, 2, 7, 6, 10)$.

Find the predictive distribution for the number of flaws in another fibre of length 60 meters.

5. Find the asymptotic posterior distributions for θ for the model where:

(a) $x_1, \dots, x_n$ are n *iid* samples from

$$f(x|\theta) = \theta^{x-1}(1-\theta) \quad x = 1, 2, \dots$$

(b) $x_1, \dots, x_n$ are n *iid* samples from

$$f(x|\theta) = \frac{\exp(-\theta)\theta^x}{x!} \quad x = 0, 1, \dots$$

6. Suppose x has the $\text{Pareto}(a, b)$ distribution, for known a , where

$$f(x|b) = ba^b x^{-b-1} \quad \text{for } x > a.$$

Find the asymptotic posterior distribution for b .

7. Suppose that $x \sim \text{Binomial}(n, \theta)$ with conjugate prior $\theta \sim \text{Beta}(a, b)$. Then we know that

$$\theta|x \sim \text{Beta}(a+x, b+n-x).$$

Now suppose we intend to make N further observations. Let y be the number of successes, so that $y|\theta \sim \text{Binomial}(N, \theta)$. Hence

$$f(y|\theta) = \binom{N}{y} \theta^y (1-\theta)^{N-y}.$$

Derive the predictive distribution for $y|x$.